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Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

EFFICIENTNET_V2_S BASED TRANSFER LEARNING FRAMEWORK FOR INDIAN CATTLE BREED RECOGNITION

A PROJECT REPORT

Submitted by

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Under the guidance of,

Mr. JOHN BENNET JOHNSON

in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

INFORMATION SCIENCE AND ENGINEERING

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report **"EfficientNet_V2_S Based Transfer Learning Framework For Indian Catte Breed Recognition"** is a Bonafide work of **"ADITYA V S (20221ISE0032), CHIANTHAN T C NADIG (20221ISE0062)"**, who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in INFORMATION SCIENCE AND ENGINEERING, during 2025-26.

22/4

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DECLARATION

We, **ADITYA V S and CHIRANTHAN T C NADIG,** a final-year B.Tech student in Information Science and Engineering at Presidency University, Bengaluru, hereby declare that the project titled **"EfficientNet_V2_S Based Transfer Learning Framework for Indian Cattle Breed Recognition,"** has been carried out by me independently under the prescribed guidelines.

This project is submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Information Science and Engineering** during the academic year 2025–2026.

We further declare that the work presented in this project report is original and has not been submitted, either in part or in full, for the award of any Degree or Diploma to any other University or Institution.

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ADITYA V S

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ABSTRACT

The rapid advancement of deep learning and computer vision has significantly improved automated image-based classification systems, particularly in agriculture and livestock management. In India, where livestock plays a crucial role in the economy and rural livelihoods, accurate identification of cattle and buffalo breeds is essential for enhancing dairy productivity, selective breeding, and disease management. However, traditional breed identification methods rely on manual inspection and expert knowledge, making them time-consuming, subjective, and often inaccessible to small-scale farmers.

This project, titled *"EfficientNet_V2_S Based Transfer Learning Framework for Indian Cattle Breed Recognition,"* proposes an intelligent and automated system for classifying cattle and buffalo breeds using images. The approach utilizes Transfer Learning with the EfficientNet V2 architecture, enabling high accuracy and efficient performance even with limited datasets.

The model is trained on curated datasets of Indian cattle breeds using preprocessing, normalization, and data augmentation techniques to improve robustness and generalization. A two-stage training strategy is adopted, where classifier layers are trained initially, followed by fine-tuning of deeper layers to capture breed-specific features such as body structure, horn shape, skin texture, and color patterns. Model performance is evaluated using accuracy and confusion matrix analysis to ensure reliability.

To enhance usability, the trained model is deployed as a Streamlit-based web application, allowing users to upload images and receive real-time predictions. The system displays the top predicted breeds along with their probability scores, providing insight into prediction confidence. The application ensures fast and efficient inference without requiring technical expertise.

Overall, the proposed system addresses key challenges associated with manual breed identification by providing an accessible, scalable, and efficient solution. It supports farmers, researchers, and agricultural stakeholders by improving decision-making, reducing human effort, and promoting the adoption of smart technologies in livestock management, thereby contributing to the advancement of precision agriculture and digital transformation in the livestock sector.